DTC II -→ Transformada discreta del Coseno

Partiendo de una matriz A de 8x 8 de muestra.

$$A = \begin{pmatrix} 200 & 200 & 200 & 200 & 50 & 50 & 50 & 50 \\ 200 & 200 & 200 & 200 & 50 & 50 & 50 & 50 \\ 200 & 200 & 200 & 200 & 50 & 50 & 50 & 50 \\ 200 & 200 & 200 & 200 & 50 & 50 & 50 & 50 \\ 100 & 100 & 100 & 100 & 20 & 20 & 20 & 20 \\ 100 & 100 & 100 & 100 & 20 & 20 & 20 & 20 \\ 100 & 100 & 100 & 100 & 20 & 20 & 20 & 20 \\ 100 & 100 & 100 & 100 & 20 & 20 & 20 & 20 \end{pmatrix}$$

Bajo la fórmula exacta de en radianes para la trasformada discreta del coseno ortogonal DCT – II es:

$$d_{i,j} = \begin{cases} \frac{1}{\sqrt{8}} \approx 0.35355 & \text{si } i = 0 \\ \sqrt{\frac{2}{8}} \cos\left(\frac{(2j+1) \cdot i \cdot \pi}{16}\right) & \text{si } i > 0 \end{cases}$$

Con esto vamos a calcular la Matriz D8, que seria la matriz tranformada

Si i = fila y l=j → columna:

Para i = 0 y todas las columnas:

$$D_8(0,l) = \left(\frac{\sqrt{2}}{4} \quad \frac{\sqrt{2}}{4} \quad \frac{\sqrt{2}}{4} \quad \frac{\sqrt{2}}{4} \quad \frac{\sqrt{2}}{4} \quad \frac{\sqrt{2}}{4} \quad \frac{\sqrt{2}}{4} \quad \frac{\sqrt{2}}{4}\right)$$

$$D_8(0,l) = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4} = 0.3535533906$$

Para i = 1 y todas las columnas

$$D_8(1,l) = \sqrt{\frac{2}{8}} \cos\left(\frac{\pi(1)(2l+1)}{16}\right) = \sqrt{\frac{2}{8}} \cos\left(\frac{\pi(2l+1)}{16}\right)$$

$$l = 0\text{: } D_8(1,0) = \sqrt{\frac{2}{8}} \cos\left(\frac{\pi(2(0)+1)}{16}\right) = \sqrt{\frac{2}{8}} \cos\left(\frac{\pi}{16}\right) = 0.4903926402$$

$$l = 1\text{: } D_8(1,1) = \sqrt{\frac{2}{8}} \cos\left(\frac{\pi(2(1)+1)}{16}\right) = \sqrt{\frac{2}{8}} \cos\left(\frac{3\pi}{16}\right) = 0.4157348062$$

$$l = 2\text{: } D_8(1,2) = \sqrt{\frac{2}{8}} \cos\left(\frac{\pi(2(2)+1)}{16}\right) = \sqrt{\frac{2}{8}} \cos\left(\frac{5\pi}{16}\right) = 0.2777851165$$

$$l = 3\text{: } D_8(1,3) = \sqrt{\frac{2}{8}} \cos\left(\frac{\pi(2(3)+1)}{16}\right) = \sqrt{\frac{2}{8}} \cos\left(\frac{7\pi}{16}\right) = 0.09754516105$$

$$l = 4:\ D_8(1,4) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(2(4)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{9\pi}{16}\right) = -0.09754516101$$

$$l = 5:\ D_8(1,5) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(2(5)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{11\pi}{16}\right) = -0.2777851165$$

$$l = 6:\ D_8(1,6) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(2(6)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{13\pi}{16}\right) = -0.4157348062$$

$$l = 7:\ D_8(1,7) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(2(7)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{15\pi}{16}\right) = -0.4903926402$$

$$D_8(1,:) = \begin{bmatrix} 0.4903926402 \\ 0.4157348062 \\ 0.2777851165 \\ 0.09754516105 \\ -0.09754516101 \\ -0.2777851165 \\ -0.4157348062 \\ -0.4903926402 \end{bmatrix}$$

Para i = 2 y en j todas las columnas

$$D_8(2,l) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(2)(2l+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(2l+1)}{16}\right)$$

$$l = 0:\ D_8(2,0) = \sqrt{\frac{2}{8}}\cos\left(\frac{2\pi(2(0)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi}{8}\right) = 0.4619397663$$

$$l = 1:\ D_8(2,1) = \sqrt{\frac{2}{8}}\cos\left(\frac{2\pi(2(1)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi}{8}\right) = 0.1913417162$$

$$l = 2:\ D_8(2,2) = \sqrt{\frac{2}{8}}\cos\left(\frac{2\pi(2(2)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi}{8}\right) = -0.191341716$$

$$l = 3:\ D_8(2,3) = \sqrt{\frac{2}{8}}\cos\left(\frac{2\pi(2(3)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi}{8}\right) = -0.4619397663$$

$$l = 4:\ D_8(2,4) = \sqrt{\frac{2}{8}}\cos\left(\frac{2\pi(2(4)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{9\pi}{8}\right) = -0.4619397663$$

$$l = 5:\ D_8(2,5) = \sqrt{\frac{2}{8}}\cos\left(\frac{2\pi(2(5)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{11\pi}{18}\right) = -0.1913417162$$

$$l = 6:\ D_8(2,6) = \sqrt{\frac{2}{8}}\cos\left(\frac{2\pi(2(6)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{13\pi}{8}\right) = 0.1913417162$$

$$l = 7:\ D_8(2,7) = \sqrt{\frac{2}{8}}\cos\left(\frac{2\pi(2(7)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{15\pi}{8}\right) = 0.4619397663$$

$$D_8(2,:) = \begin{bmatrix} 0.4619397663 \\ 0.1913417162 \\ -0.191341716 \\ -0.4619397663 \\ -0.4619397663 \\ -0.1913417162 \\ 0.1913417162 \\ 0.4619397663 \end{bmatrix}$$

Para i = 3 y en j todas las columnas

$$D_8(3,l) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(3)(2l+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(2l+1)}{16}\right)$$

$$l = 0:\ D_8(3,0) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi(2(0)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi}{16}\right) = 0.4157348062$$

$$l = 1:\ D_8(3,1) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi(2(1)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{9\pi}{16}\right) = -0.09754516101$$

$$l = 2:\ D_8(3,2) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi(2(2)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{15\pi}{16}\right) = -0.4903926402$$

$$l = 3:\ D_8(3,3) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi(2(3)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{21\pi}{16}\right) = -0.2777851165$$

$$l = 4:\ D_8(3,4) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi(2(4)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{27\pi}{16}\right) = 0.2777851165$$

$$l = 5:\ D_8(3,5) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi(2(5)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{33\pi}{16}\right) = 0.4903926402$$

$$l = 6:\ D_8(3,6) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi(2(6)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{39\pi}{16}\right) = 0.09754516101$$

$$l = 7:\ D_8(3,7) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi(2(7)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{45\pi}{16}\right) = -0.4157348062$$

$$D_8(3,:) = \begin{bmatrix} 0.4157348062 \\ -0.09754516101 \\ -0.4903926402 \\ -0.27778511653 \\ 0.2777851165 \\ 0.4903926402 \\ 0.09754516101 \\ -0.4157348062 \end{bmatrix}$$

Para i = 4 y en j todas las columnas

$$D_8(4,l) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(4)(2l+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2l+1)}{16}\right)$$

$$l = 0:\ D_8(4,0) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2(0)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi}{4}\right) = 0.3535533906$$

$$l = 1:\ D_8(4,1) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2(1)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{3\pi}{4}\right) = -0.3535533906$$

$$l = 2:\ D_8(4,2) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2(2)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi}{4}\right) = -0.3535533906$$

$$l = 3:\ D_8(4,3) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2(3)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi}{4}\right) = 0.3535533906$$

$$l = 4:\ D_8(4,4) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2(4)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{9\pi}{4}\right) = 0.3535533906$$

$$l = 5:\ D_8(4,5) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2(5)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{11\pi}{4}\right) = -0.3535533906$$

$$l = 6:\ D_8(4,6) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2(6)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{13\pi}{4}\right) = -0.3535533906$$

$$l = 7:\ D_8(4,7) = \sqrt{\frac{2}{8}}\cos\left(\frac{4\pi(2(7)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{15\pi}{4}\right) = 0.3535533906$$

$$D_8(4,:) = \begin{bmatrix} 0.3535533906 \\ -0.3535533906 \\ -0.3535533906 \\ 0.3535533906 \\ 0.3535533906 \\ -0.3535533906 \\ -0.3535533906 \\ 0.3535533906 \end{bmatrix}$$

Para i = 5 y en j todas las columnas

$$D_8(5,l) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(5)(2l+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2l+1)}{16}\right)$$

$$l = 0:\ D_8(5,0) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2(0)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi}{16}\right) = 0.2777851165$$

$$l = 1:\ D_8(5,1) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2(1)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{15\pi}{16}\right) = -0.4903926402$$

$$l = 2:\ D_8(5,2) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2(2)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{25\pi}{16}\right) = 0.09754516101$$

$$l = 3:\ D_8(5,3) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2(3)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{35\pi}{16}\right) = 0.4157348062$$

$$l = 4:\ D_8(5,4) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2(4)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{45\pi}{16}\right) = -0.4157348062$$

$$l = 5:\ D_8(5,5) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2(5)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{55\pi}{16}\right) = -0.09754516101$$

$$l = 6:\ D_8(5,6) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2(6)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{65\pi}{16}\right) = 0.4903926402$$

$$l = 7:\ D_8(5,7) = \sqrt{\frac{2}{8}}\cos\left(\frac{5\pi(2(7)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{75\pi}{16}\right) = -0.2777851165$$

$$D_8(5,:) = \begin{bmatrix} 0.2777851165 \\ -0.4903926402 \\ 0.09754516101 \\ 0.4157348062 \\ -0.4157348062 \\ -0.09754516101 \\ 0.4903926402 \\ -0.2777851165 \end{bmatrix}$$

Para i = 6 y en j todas las columnas

$$D_8(6,l) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(6)(2l+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2l+1)}{16}\right)$$

$$l = 0:\ D_8(6,0) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2(0)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi}{16}\right) = 0.1913417162$$

$$l = 1:\ D_8(6,1) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2(1)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{18\pi}{16}\right) = -0.4619397663$$

$$l = 2:\ D_8(6,2) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2(2)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{30\pi}{16}\right) = 0.4619397663$$

$$l = 3:\ D_8(6,3) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2(3)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{42\pi}{16}\right) = -0.1913417162$$

$$l = 4:\ D_8(6,4) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2(4)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{54\pi}{16}\right) = -0.1913417162$$

$$l = 5:\ D_8(6,5) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2(5)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{66\pi}{16}\right) = 0.4619397663$$

$$l = 6:\ D_8(6,6) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2(6)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{78\pi}{16}\right) = -0.4619397663$$

$$l = 7:\ D_8(6,7) = \sqrt{\frac{2}{8}}\cos\left(\frac{6\pi(2(7)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{90\pi}{16}\right) = 0.1913417162$$

$$D_8(6,:) = \begin{bmatrix} 0.1913417162 \\ -0.4619397663 \\ 0.4619397663 \\ -0.1913417162 \\ -0.1913417162 \\ 0.4619397663 \\ -0.4619397663 \\ 0.1913417162 \end{bmatrix}$$

Para i = 7 y en j todas las columnas

$$D_8(7,l) = \sqrt{\frac{2}{8}}\cos\left(\frac{\pi(7)(2l+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2l+1)}{16}\right)$$

$$l = 0:\ D_8(7,0) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2(0)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi}{16}\right) = 0.09754516101$$

$$l = 1:\ D_8(7,1) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2(1)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{21\pi}{16}\right) = -0.277785116$$

$$l = 2:\ D_8(7,2) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2(2)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{35\pi}{16}\right) = 0.4157348062$$

$$l = 3:\ D_8(7,3) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2(3)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{49\pi}{16}\right) = -0.4903926402$$

$$l = 4:\ D_8(7,4) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2(4)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{63\pi}{16}\right) = 0.4903926402$$

$$l = 5:\ D_8(7,5) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2(5)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{77\pi}{16}\right) = -0.4157348062$$

$$l = 6:\ D_8(7,6) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2(6)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{91\pi}{16}\right) = 0.2777851165$$

$$l = 7:\ D_8(7,7) = \sqrt{\frac{2}{8}}\cos\left(\frac{7\pi(2(7)+1)}{16}\right) = \sqrt{\frac{2}{8}}\cos\left(\frac{105\pi}{16}\right) = -0.09754516101$$

$$D_8(7,:) = \begin{bmatrix} 0.09754516101 \\ -0.277785116 \\ 0.415734806 \\ -0.4903926402 \\ 0.4903926402 \\ -0.4157348062 \\ 0.2777851165 \\ -0.09754516101 \end{bmatrix}$$

La matriz D8 que estará compuesta por las filas anteriormente calculadas

$$D_8 = \begin{pmatrix} 0.35355 & 0.35355 & 0.35355 & 0.35355 & 0.35355 & 0.35355 & 0.35355 & 0.35355 \\ 0.49039 & 0.41573 & 0.27779 & 0.09755 & -0.09755 & -0.27779 & -0.41573 & -0.49039 \\ 0.46194 & 0.19134 & -0.19134 & -0.46194 & -0.46194 & -0.19134 & 0.19134 & 0.46194 \\ 0.41573 & -0.09755 & -0.49039 & -0.27779 & 0.27779 & 0.49039 & 0.09755 & -0.41573 \\ 0.35355 & -0.35355 & -0.35355 & 0.35355 & 0.35355 & -0.35355 & -0.35355 & 0.35355 \\ 0.27779 & -0.49039 & 0.09755 & 0.41573 & -0.41573 & -0.09755 & 0.49039 & -0.27779 \\ 0.19134 & -0.46194 & 0.46194 & -0.19134 & -0.19134 & 0.46194 & -0.46194 & 0.19134 \\ 0.09755 & -0.27779 & 0.41573 & -0.49039 & 0.49039 & -0.41573 & 0.27779 & -0.09755 \end{pmatrix}$$

$$C = D_8 * A * D_8^T$$

$$D_8 * D_8^T = I$$

Queremos transformar la imagen original, que está representada por píxeles, en una representación basada en frecuencias espaciales. Para eso calculamos C.

$$C = \begin{pmatrix} 0.35355 & \cdots & 0.3555 \\ \vdots & \ddots & \vdots \\ 0.09755 & \cdots & -0.09755 \end{pmatrix} \begin{pmatrix} 200 & \cdots & 50 \\ \vdots & \ddots & \vdots \\ 100 & \cdots & 20 \end{pmatrix} \begin{pmatrix} 0.35355 & \cdots & 0.09755 \\ \vdots & \ddots & \vdots \\ 0.35355 & \cdots & -0.09755 \end{pmatrix}$$

$$\begin{pmatrix} 0.35355 & 0.35355 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 0.35355 \\ 0.49039 & 0.41573 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & -0.49039 \\ 0.46194 & -0.09755 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 0.46194 \\ 0.41573 & -0.35355 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & -0.41573 \\ 0.35355 & -0.49039 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 0.35355 \\ \ldots & \ldots & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & \cdot\cdot \\ 0.09755 & -0.27778 & \cdot\cdot & \cdot\cdot & \cdot\cdot & \cdot\cdot & -0.09755 \end{pmatrix}_{8x8} \begin{pmatrix} 200 & \cdots & 50 \\ \vdots & \ddots & \vdots \\ 100 & \cdots & 20 \end{pmatrix}_{8x8} \begin{pmatrix} 0.35355 & \cdots & 0.09755 \\ \vdots & \ddots & \vdots \\ 0.35355 & \cdots & -0.09755 \end{pmatrix}_{8x8}$$

Verificamos ortogonalidad y ortonormalidad por filas de D8.

$$\sum_{n=0}^{7} (f_{0,n})^2 = \sum_{n=0}^{7} c_n^2 \rightarrow \sum_{n=0}^{7} \left(\frac{1}{\sqrt{8}}\right)^2 = \left(\frac{1}{\sqrt{8}}\right)^2 + \left(\frac{1}{\sqrt{8}}\right)^2 + \cdots + \left(\frac{1}{\sqrt{8}}\right)^2 = 1$$

Sus valores teóricos son

$$\frac{1}{2} cos\left(\frac{(2n+1)\pi}{8}\right) \rightarrow \sum_{n=0}^{7} \left[\frac{1}{2}\cos\left(\frac{(2n+1)\pi}{8}\right)\right]^2$$

$$D_8 * A = \begin{pmatrix} 0.35355 & 0.35355 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 0.35355 \\ 0.49039 & 0.41573 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & -0.49039 \\ 0.46194 & -0.09755 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 0.46194 \\ 0.41573 & -0.35355 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & -0.41573 \\ 0.35355 & -0.49039 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 0.35355 \\ \ldots & \ldots & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & \ldots \\ 0.09755 & -0.27778 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & -0.09755 \end{pmatrix} x \begin{pmatrix} 200 & 200 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 50 \\ 200 & 200 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 50 \\ 200 & 200 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 50 \\ 200 & 200 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 50 \\ 100 & 100 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 20 \\ \ldots & \ldots & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & \ldots \\ 100 & 100 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 20 \end{pmatrix}$$

$$A = \begin{pmatrix} 200 & 200 & 200 & 200 & 50 & 50 & 50 & 50 \\ 200 & 200 & 200 & 200 & 50 & 50 & 50 & 50 \\ 200 & 200 & 200 & 200 & 50 & 50 & 50 & 50 \\ 200 & 200 & 200 & 200 & 50 & 50 & 50 & 50 \\ 100 & 100 & 100 & 100 & 20 & 20 & 20 & 20 \\ 100 & 100 & 100 & 100 & 20 & 20 & 20 & 20 \\ 100 & 100 & 100 & 100 & 20 & 20 & 20 & 20 \\ 100 & 100 & 100 & 100 & 20 & 20 & 20 & 20 \end{pmatrix}$$

Para cada fila de la matriz D_8, la sumatoria del producto punto es:

Fila D8_1 x las 4 columnas: donde n y m es fila y columna respectivamente

Fn*Cm = $\mathbf{200 \cdot (d_1 + d_2 + d_3 + d_4) + 100 \cdot (d_5 + d_6 + d_7 + d_8)}$

$$(D_8 \cdot A)_{i,k} = \sum_{j=1}^{8} d_{i,j} \cdot a_{j,k}$$

Para las primeras 4 columnas de la Matriz A

$$(D_8 \cdot A)_{i,k} = 200 \sum_{j=1}^{4} d_{i,j} \; + \; 100 \sum_{j=5}^{8} d_{i,j}$$

Para las ultimas 4 columnas de la Matriz A:

$$(D_8 \cdot A)_{i,k} = 50 \sum_{j=1}^{4} d_{i,j} \; + \; 20 \sum_{j=5}^{8} d_{i,j}$$

Fila n=1 y m={1,2,3,4}

$$= 200 \cdot (0.35355 * 4) + 100 \cdot (0.35355 * 4) = 424.26$$

Fila n=1 y m={5,6,7,8}

$$= 50 \cdot (0.35355 * 4) + 20 \cdot (0.35355 * 4) = 98.994$$

Fila n=2 y m={1,2,3,4}

$$= 200 \cdot (0.49039 + 0.41573 + 0.27779 + 0.09755) + 100 \cdot (-0.09755 - 0.27779 - 0.41573 - 0.49039) = 128.146$$

Fila n=2 y m={5,6,7,8}

$$= 50 \cdot (0.49039 + 0.41573 + 0.27779 + 0.09755) + 20 \cdot (-0.09755 - 0.27779 - 0.41573 - 0.49039) = 38.4438$$

Fila n=3 y m={1,2,3,4}

$$= 200 \cdot (0.46194 + 0.19134 - 0.19134 - 0.46194) + 100 \cdot (-0.46194 - 0.19134 + 0.19134 + 0.46194) = 0$$

Fila n=3 y m={5,6,7,8}

$$= 50 \cdot (0.46194 + 0.19134 - 0.19134 - 0.46194) + 20 \cdot (-0.46194 - 0.19134 + 0.19134 + 0.46194) = 0$$

Fila n=4 y m={1,2,3,4}:

$$= 200 \cdot (0.41573 - 0.09755 - 0.49039 - 0.27779) + 100 \cdot (0.27779 + 0.49039 + 0.09755 - 0.41573) = -45$$

Fila n=4 y m={5,6,7,8}

$$50 \cdot (0.41573 - 0.09755 - 0.49039 - 0.27779) + 20 \cdot (0.27779 + 0.49039 + 0.09755 - 0.41573) = -13.5$$

Fila n=5 y m={1,2,3,4}:

$$= 200 \cdot (0.35355 - 0.35355 - 0.35355 + 0.35355) + 100 \cdot (0.35355 - 0.35355 - 0.35355 + 0.35355) = 0.0$$

Fila n=5 y m={5,6,7,8}

$$= 50 \cdot (0.35355 - 0.35355 - 0.35355 + 0.35355) + 20 \cdot (0.35355 - 0.35355 - 0.35355 + 0.35355) = 0.0$$

Fila n=6 y m={1,2,3,4}:

$$= 200 \cdot (0.27779 - 0.49039 + 0.09755 + 0.41573) + 100 \cdot (-0.41573 - 0.09755 + 0.49039 - 0.27779) = 30.06800$$

Fila n=6 y m={5,6,7,8}

$$= 50 \cdot (0.27779 - 0.49039 + 0.09755 + 0.41573) + 20$$
$$\cdot (-0.41573 - 0.09755 + 0.49039 - 0.27779) = \mathbf{9.02040}$$

Fila n=7 y m={1,2,3,4}:

$$= 200 \cdot (0.19134 - 0.46194 + 0.46194 - 0.19134) + 100$$
$$\cdot (-0.19134 + 0.46194 - 0.46194 + 0.19134) = 0.0$$

Fila n=7 y m={5,6,7,8}

$$= 50 \cdot (0.19134 - 0.46194 + 0.46194 - 0.19134) + 20$$
$$\cdot (-0.19134 + 0.46194 - 0.46194 + 0.19134) = 0.0$$

Fila n=8 y m={1,2,3,4}:

$$= 200 \cdot (0.09755 - 0.27778 + 0.41573 - 0.49039) + 100$$
$$\cdot (-0.41573 + 0.27779 + 0.27779 - 0.09754) = -25.49$$

Fila n=8 y m={5,6,7,8}

$$= 50 \cdot (0.09755 - 0.27778 + 0.41573 - 0.49039) + 20$$
$$\cdot (-0.41573 + 0.27779 + 0.27779 - 0.09754) = -7.647$$

$$B = \begin{pmatrix} 424.260 & 424.26 & 424.26 & 424.26 & 98.994 & 98.994 & 98.994 & 98.994 \\ 128.146 & 128.146 & 128.146 & 128.146 & 38.4438 & 38.4438 & 38.44380 & 38.4438 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ -45.0 & -45.0 & -45.0 & -45.0 & -13.5 & -13.5 & -13.5 & -13.5 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 30.068 & 30.068 & 30.068 & 30.068 & 9.0204 & 9.0204 & 9.0204 & 9.0204 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ -25.49 & -25.49 & -25.49 & -25.49 & -7.647 & -7.647 & -7.647 & -7.647 \end{pmatrix}$$

$$(D_8 * A) * D_8^T = \begin{pmatrix} 424.26 & 424.26 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 98.994 \\ 128.146 & 128.146 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 38.4438 \\ 0.00 & 0.00 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 0.00 \\ -45.00 & -45.00 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & -13.50 \\ 0.00 & 0.00 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 0.00 \\ \ldots & \ldots & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & \ldots \\ -25.49 & -25.49 & \ldots & \cdot\cdot & \cdot\cdot & \cdot\cdot & -7.647 \end{pmatrix} \begin{pmatrix} 0.35355 & \ldots & \ldots & 0.09755 \\ 0.35355 & \ldots & \ldots & -0.27778 \\ 0.35355 & \ldots & \cdots & 0.41573 \\ 0.35355 & \cdots & \ldots & -0.49039 \\ 0.35355 & \ldots & \vdots & 0.49039 \\ \cdots & \vdots & & \cdots \\ 0.35355 & \vdots & \ldots & -0.09755 \end{pmatrix}$$

$$(B \cdot D_8^T)_{i,j} = \sum_{k=1}^{8} b_{i,k} \cdot d_{j,k}$$

$$C = \begin{pmatrix} 739.98581 & 416.81537 & 0.0 & -146.36970 & 0.0 & 97.80098 & 0.0 & -82.91030 \\ 235.59130 & 114.94978 & 0.0 & -40.36599 & 0.0 & 26.97166 & 0.0 & -22.86509 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ -82.73070 & -40.36599 & 0.0 & 14.17500 & 0.0 & -9.47142 & 0.0 & 8.02935 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 55.27882 & 26.97166 & 0.0 & -9.47142 & 0.0 & 6.32859 & 0.0 & -5.36503 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ -46.86235 & -22.86509 & 0.0 & 8.02935 & 0.0 & -5.36503 & 0.0 & 4.54818 \end{pmatrix}$$

Elegimos el 25% de la coeficientes totales de la matriz C de coeficientes: igual a 16 en valor absoluto.

$$|C| = \begin{pmatrix} 739.98581 & 416.81537 & 0.0 & 146.36970 & 0.0 & 97.80098 & 0.0 & 82.91030 \\ 235.59130 & 114.94978 & 0.0 & 40.36599 & 0.0 & 26.97166 & 0.0 & 22.86509 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 82.73070 & 40.36599 & 0.0 & 14.17500 & 0.0 & 9.47142 & 0.0 & 8.02935 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 55.27882 & 26.97166 & 0.0 & 9.47142 & 0.0 & 6.32859 & 0.0 & 5.36503 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 46.86235 & 22.86509 & 0.0 & 8.02935 & 0.0 & 5.36503 & 0.0 & 4.54818 \end{pmatrix}$$

$$\mathrm{m}_{ij} = \begin{cases} m_{ij}, & \text{si } m_{ij} \text{ está entre los 16 seleccionados} \\ 0, & \text{en caso contrario} \end{cases}$$

$$M = \begin{pmatrix} 739.98581 & 416.81537 & 0.0 & -146.36970 & 0.0 & 97.80098 & 0.0 & -82.91030 \\ 235.59130 & 114.94978 & 0.0 & -40.36599 & 0.0 & 26.97166 & 0.0 & -22.86509 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ -82.73070 & -40.36599 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 55.27882 & 26.97166 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ -46.86235 & -22.86509 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{pmatrix}$$

$$\tilde{A} = D_8^T * \mathrm{M} * D_8$$

Como D8 es ortonormaal su inversa es igual a su transpuesta.

$$D_8^{-1} = D_8^T$$

$$E_{i,j} = \sum_{k=1}^{8} (D_8^T)_{i,k} \cdot M_{k,j} = \sum_{k=1}^{8} d_{k,i} \cdot M_{k,j}$$

$$E = \begin{pmatrix} 353.54444 & 192.21591 & 0.0 & -71.54409 & 0.0 & 47.80417 & 0.0 & -40.52575 \\ 353.54444 & 192.21591 & 0.0 & -68.53036 & 0.0 & 45.79046 & 0.0 & -38.81864 \\ 353.54756 & 192.21743 & 0.0 & -62.96228 & 0.0 & 42.06999 & 0.0 & -35.66463 \\ 353.54756 & 192.21743 & 0.0 & -55.68671 & 0.0 & 37.20862 & 0.0 & -31.54343 \\ 169.69640 & 102.51271 & 0.0 & -47.81131 & 0.0 & 31.94645 & 0.0 & -27.08245 \\ 169.69640 & 102.51271 & 0.0 & -40.53574 & 0.0 & 27.08508 & 0.0 & -22.96124 \\ 169.69952 & 102.51424 & 0.0 & -34.96765 & 0.0 & 23.36461 & 0.0 & -19.80723 \\ 169.69952 & 102.51424 & 0.0 & -31.95393 & 0.0 & 21.35091 & 0.0 & -18.10013 \end{pmatrix}$$

$$(E \cdot D_8)_{i,j} = \sum_{k=1}^{8} e_{i,k} \cdot d_{k,j}$$

$$\tilde{A} = \begin{pmatrix} 198.83961 & 199.69965 & 201.29133 & 203.36758 & 46.62370 & 48.69995 & 50.29163 & 51.15167 \\ 199.69965 & 199.91894 & 200.32669 & 200.85609 & 49.13518 & 49.66459 & 50.07234 & 50.29163 \\ 201.29048 & 200.32584 & 198.54596 & 196.21718 & 53.77630 & 51.44752 & 49.66764 & 48.70300 \\ 203.36673 & 200.85524 & 196.21718 & 190.15408 & 59.83940 & 53.77630 & 49.13824 & 46.62675 \\ 96.62329 & 99.13478 & 103.77675 & 109.83984 & 10.15249 & 16.21558 & 20.85755 & 23.36903 \\ 98.69955 & 99.66419 & 101.44796 & 103.77675 & 16.21558 & 18.54436 & 20.32814 & 21.29278 \\ 100.29038 & 100.07108 & 99.66724 & 99.13783 & 20.85670 & 20.32729 & 19.92345 & 19.70415 \\ 101.15042 & 100.29038 & 98.70260 & 96.62634 & 23.36819 & 21.29193 & 19.70415 & 18.84412 \end{pmatrix}$$

Calculamos la matriz de los errores W.

$$W = A - \tilde{A}$$

$$W = \begin{pmatrix} 200 & 200 & \dots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 50 \\ 200 & 200 & \dots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 50 \\ 200 & 200 & \dots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 50 \\ 200 & 200 & \dots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 50 \\ 100 & 100 & \dots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 20 \\ \dots & \dots & \dots & \cdot\cdot & \cdot\cdot & \cdot\cdot & \dots \\ 100 & 100 & \dots & \cdot\cdot & \cdot\cdot & \cdot\cdot & 20 \end{pmatrix} - \begin{pmatrix} 198.83961 & \dots & \dots & 51.15167 \\ 199.69965 & \dots & \dots & 50.29163 \\ 201.29048 & \dots & \cdots & 48.70300 \\ 203.36673 & \cdots & \dots & 46.62675 \\ 96.62329 & \dots & \vdots & 23.36903 \\ \dots & \vdots & \vdots & \dots \\ 101.15042 & \vdots & \dots & 18.84412 \end{pmatrix}$$

$$W = \begin{pmatrix} 1.16039 & 0.30035 & -1.29133 & -3.36758 & 3.37630 & 1.30005 & -0.29163 & -1.15167 \\ 0.30035 & 0.08106 & -0.32669 & -0.85609 & 0.86482 & 0.33541 & -0.07234 & -0.29163 \\ -1.29048 & -0.32584 & 1.45404 & 3.78282 & -3.77630 & -1.44752 & 0.33236 & 1.29700 \\ -3.36673 & -0.85524 & 3.78282 & 9.84592 & -9.83940 & -3.77630 & 0.86176 & 3.37325 \\ 3.37671 & 0.86522 & -3.77675 & -9.83984 & 9.84751 & 3.78442 & -0.85755 & -3.36903 \\ 1.30045 & 0.33581 & -1.44796 & -3.77675 & 3.78442 & 1.45564 & -0.32814 & -1.29278 \\ -0.29038 & -0.07108 & 0.33276 & 0.86217 & -0.85670 & -0.32729 & 0.07655 & 0.29585 \\ -1.15042 & -0.29038 & 1.29740 & 3.37366 & -3.36819 & -1.29193 & 0.29585 & 1.15588 \end{pmatrix}$$

Como la imagen es de 8 x 8 y tiene 64 pixeles:

$$\text{MSE} = \frac{1}{mn} \sum_{i=1}^{m} \sum_{j=1}^{n} \left(a_{ij} - \tilde{a}_{ij}\right)^2 \rightarrow \ = \frac{1}{64} \sum_{i=1}^{8} \sum_{j=1}^{8} w_{ij}^2$$

$$\sum_{i=1}^{8} \sum_{j=1}^{8} w_{i,j}^2 = 627.48065 \ \rightarrow \ \text{MSE} = \frac{627.48065}{64} \approx 9.80439$$

El MSE no está expresado en unidades de intensidad originales, sino en unidades de intensidad al cuadrado. Cuanto menor sea el MSE, menor será el error promedio de reconstrucción.

Usamos PSNR para calcular los decibeles.

$$PSNR = 10 * log_{10} * \left(\frac{MAX^2}{MSE}\right)$$

$$PSNR = 10 * log_{10} * \left(\frac{255^2}{9.80439}\right) = 10 * log_{10} * \left(\frac{65\,025}{9.80439}\right) \approx 38.21660\ dB$$